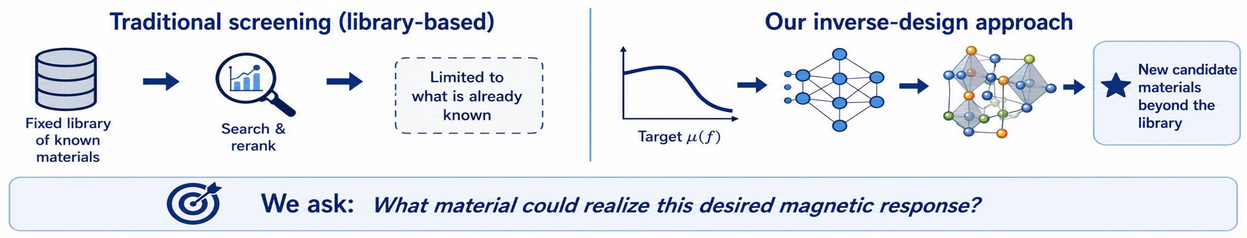
MagFlow: Inverse Design of High-Frequency Magnetic Materials

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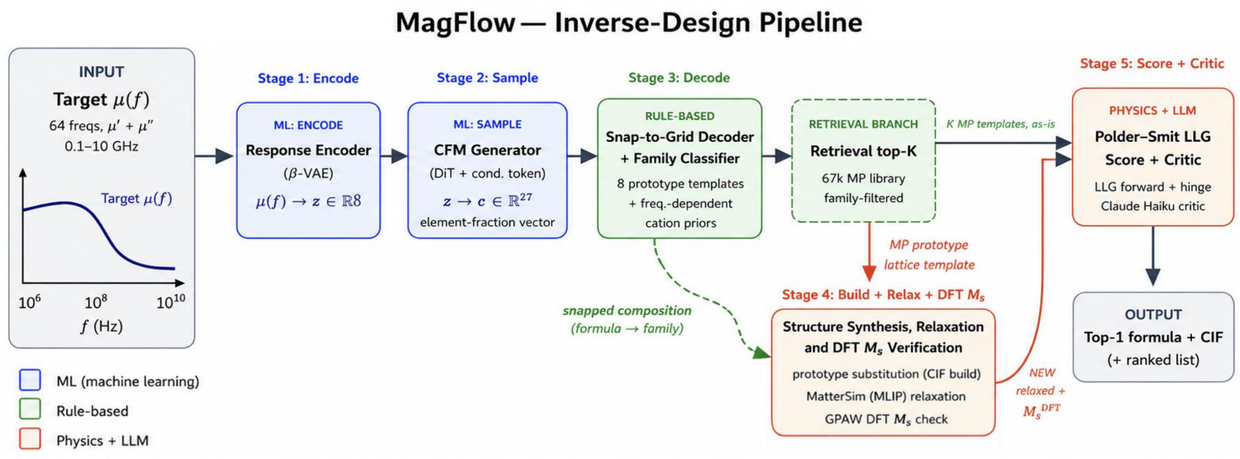
Motivation

Searching a fixed library of known materials to find candidates with the desired magnetic performance is inherently limited. To address this, we propose MagFlow, an inverse-design pipeline that **generates candidate high-frequency soft magnetic materials** from a desired permeability spectrum μ(f).



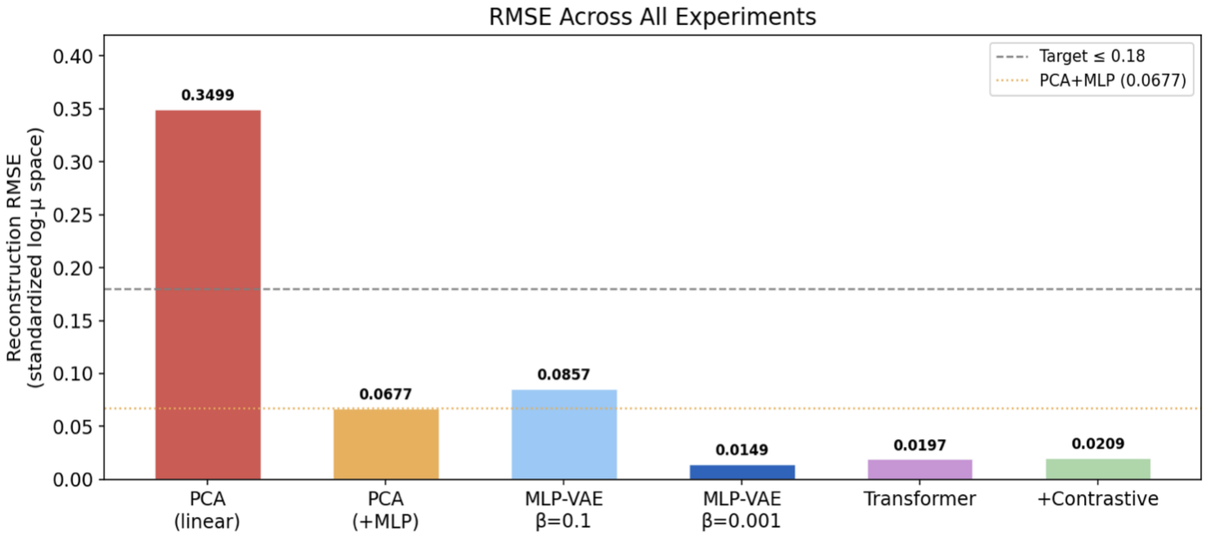
Pipeline overview

Given a target permeability spectrum $\mu(f)$, MagFlow (1) encodes the response, (2) generates candidate compositions with CFM, (3) snaps them to chemically valid formula–prototype pairs, (4) builds, relaxes, and verifies structures with MatterSim and GPAW DFT M_s check [4,5], and (5) reranks candidates with Polder–Smit LLG [2] and an LLM chemistry critic.



Stage 1 — ResponseNet Encoder (β-VAE)

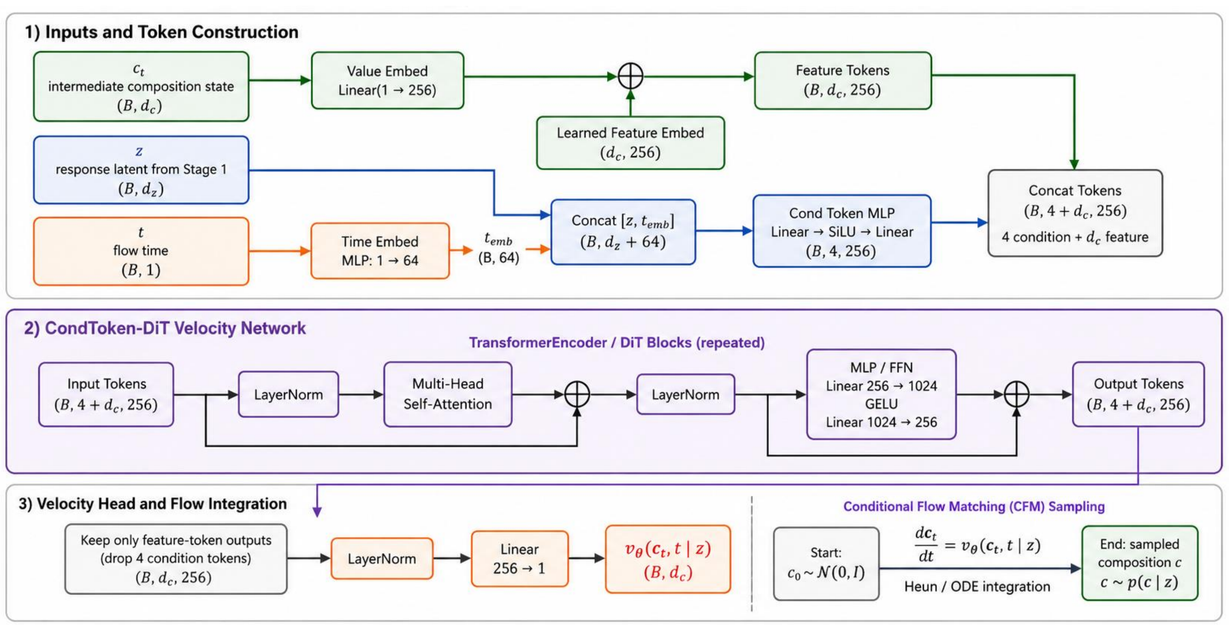
The ResponseNet Encoder compresses the 64-pt log-spaced $\mu(f)$ curve into an 8-d latent z with β-VAE training [3]. Among all configurations, $\beta = 0.001$ gives the lowest reconstruction RMSE ,with a 23× improvement over a linear PCA baseline.



Stage 2 — Conditional Flow Matching (CFM) Generator

The CFM generator learns a continuous-time velocity field $v_\theta(c_t, t|\, z)$ that maps a Gaussian prior to the conditional composition distribution $p(c \mid z)$ [3], where z is the Stage-1 response latent and c_t is the intermediate composition state.

We implement v_θ with a condition-token DiT [3]: the composition tokens are concatenated with response/time condition tokens and processed by repeated self-attention blocks. At inference, we integrate $dc_t/dt = v_\theta(c_t, t|z)$ with a 50-step Heun sampler using an EDM schedule [4] and classifier-free guidance scale = 2.



Stage 3 — Snap-to-grid Decoder + Family Priors

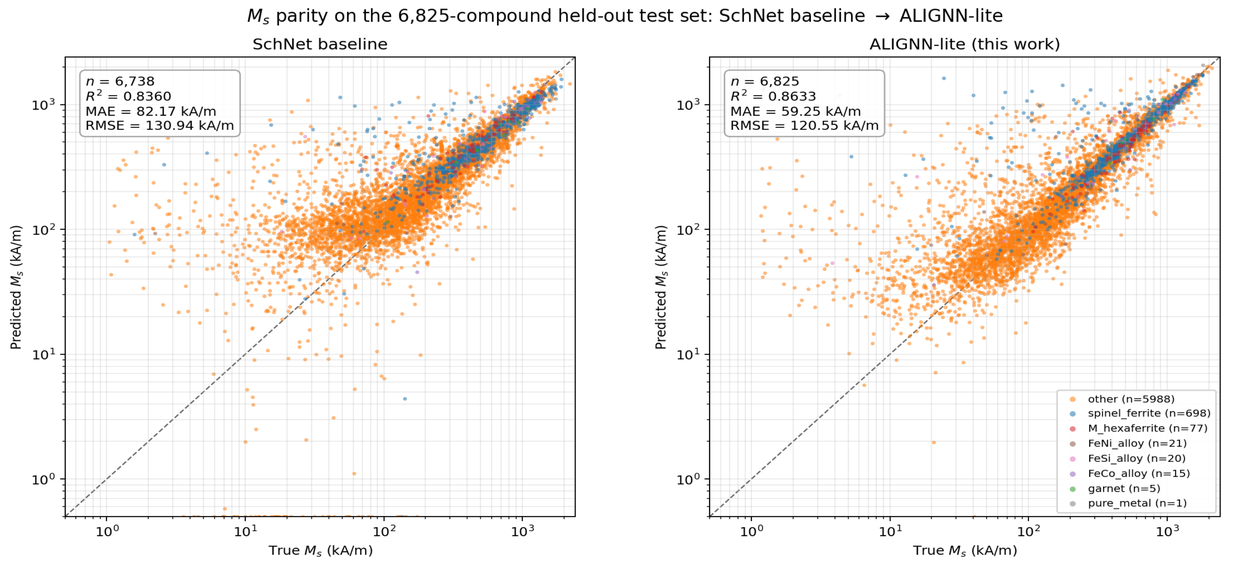
Projects continuous 27-vector onto one of eight family prototypes (spinel, M-hex, garnet, FeNi / FeCo / FeSi alloys, pure metal) via a 3-pass site-assignment algorithm. A frequency-dependent hint injects Mn, Ba, or Y as family-signature cations, reflecting the usual separation between soft low-frequency spinels and microwave hexaferrites [1]. Each family inherits literature-mean $K_1, \alpha,$ and ρ priors from ferrite, garnet, and alloy references [1,6].

Family	K_1 (kJ/m³)	α	ρ (Ω·m)	Reference
spinel_ferrite	−3.0	0.05	10^3	Smit & Wijn 1959
M_hexaferrite	+300	0.10	10^7	Pullar 2012
garnet	−0.6	10^{-4}	10^{10}	Stancil 2009
FeNi_alloy	+0.10	0.008	6×10^{-7}	Bozorth 1993
FeCo_alloy	+4.0	0.01	5×10^{-7}	Bozorth 1993
FeSi_alloy	+20	0.002	6×10^{-7}	Bozorth 1993
pure_metal	+50	0.005	10^{-7}	CRC Handbook
other (fallback)	+10	0.01	10^0	—

Stage 4 — Structure Synthesis + DFT M_s Verification

Decoded formula–prototype pairs are built into CIFs, relaxed with MatterSim, and checked by GPAW DFT [4,5]. The resulting M_s DFT is used as a sanity check. For canonical ferrites where collinear DFT can overestimate ferrimagnetic moments, literature M_s overrides are used; ALIGNN-lite backfills missing M_s values [6].

Result: improved held-out M_s prediction (R^2: 0.819→0.863; MAE: 264→195 kA/m).

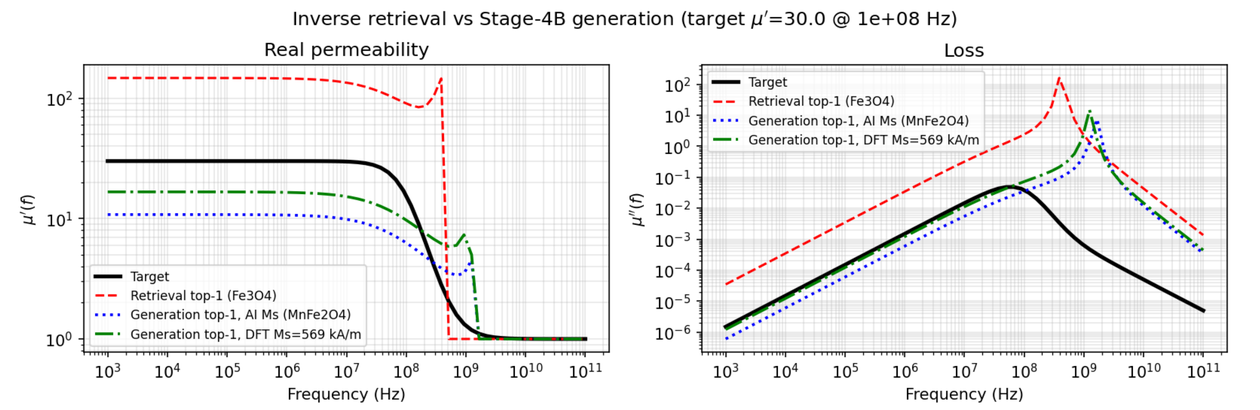


Predictor	Test R^2	MAE (kA/m)	Notes
SchNet baseline	0.819	264	architecture-matched
ALIGNN-lite (ours)	0.863	195	+5.4 % R^2, −26 % MAE
CHGNet (off-the-shelf)	0.78	310	MP-trained
MACE (off-the-shelf)	0.81	245	equivariant

Stage 5 — Polder-Smit rerank + Inverse-design demo

The Polder–Smit reranking [2] scores each candidate using (M_s, K_1, α, ρ), the key magnetic-response parameters for ferrites and high-frequency magnetic materials [1,2], over a ±2-decade target band; curves show the 100 MHz demo.

Result: MagFlow recovers plausible ferrite top-1 candidates across all three bands, with $|\text{score}| < 0.06$ in the GHz cases and +13% μ' over-spec at 100 MHz.



Target ($f, \mu', tan\,\delta$)	Generated	Prototype	M_s lit	M_s DFT	Score
100 MHz, 30, 0.005	$MnFe_2O_4$	spinel	450	569	−0.41
1 GHz, 5, 0.01	$BaFe_{12}O_{19}$	M-hex	375	1172*	−0.05
10 GHz, 2, 0.05	$BaFe_{12}O_{19}$	M-hex	375	1184*	−0.06

CONCLUSION

MagFlow demonstrates a five-stage inverse-design workflow that maps target permeability spectra to ranked magnetic-material candidates. Its main value is combining response-conditioned generation, prototype-aware decoding, MLIP+DFT structure checking, and physics-based Polder–Smit reranking. The remaining limitation is ferrimagnetic DFT bias, which is handled here by literature M_s overrides and should be addressed by magnetic-structure-aware verification in future work.

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